

PTES Report

Executive Summary:

This engagement simulated a full-scope Vulnerability Assessment and Penetration Testing (VAPT) exercise against a target system hosted in a controlled lab environment. The objective was to identify, exploit, and recommend remediation for security weaknesses. Using industry-standard tools such as Kali Linux, Metasploit, OpenVAS, and Burp Suite, multiple phases of the PTES methodology were executed, including reconnaissance, scanning, exploitation, and reporting. A critical vulnerability in the VSFTPD service allowed remote code execution, demonstrating the risk of outdated or misconfigured services.

Attack Timeline:

The assessment began with reconnaissance to identify open ports and services running on the target system. Scanning revealed an outdated VSFTPD 2.3.4 service vulnerable to a known backdoor exploit. During the exploitation phase, Metasploit's `exploit/unix/ftp/vsftpd_234_backdoor` module was used to gain unauthorized shell access to the system. Additional testing using Burp Suite identified potential weaknesses in API input handling. All findings were documented with timestamps, confirming successful compromise during the exploitation phase.

Remediation Plan:

To mitigate the identified risks, it is recommended to immediately patch or upgrade the VSFTPD service to a secure version, removing the backdoor vulnerability. Input validation should be enforced on all API endpoints to prevent injection attacks. Implementing the principle of least privilege will reduce the impact of potential breaches. Additionally, continuous vulnerability scanning using OpenVAS should be performed to identify and remediate future risks. Proper logging and monitoring mechanisms should also be deployed to detect suspicious activities in real time.

Non-Technical Briefing

This security assessment tested how vulnerable the system is to real-world cyberattacks. During the process, a critical weakness was discovered in one of the system's services, which allowed unauthorized access. This means an attacker could potentially take control of the system without proper authentication. Additional issues were identified in how the application handles user input, which could be exploited to manipulate system behaviour.

These findings highlight the importance of keeping software updated and properly configured. To reduce risk, it is recommended to apply security patches, improve input validation, and limit user permissions. Regular security scans and monitoring should also be implemented to detect and prevent future attacks. Overall, addressing these issues will significantly strengthen the system's security and reduce the likelihood of a successful cyberattack.

